A Comprehensive Framework for Credit Risk Assessment:

Bayesian Behavioral Detection, Markov Chain Migration, and Monte Carlo Simulation

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Abstract

Credit unions face significant challenges in understanding and managing credit risk at the account level, particularly after loan origination. Executives must monitor how borrower accounts migrate across delinquency stages, estimate realistic repayment expectations, and assess potential losses under uncertain economic conditions such as rising unemployment or recession. Managing large volumes of diverse loan data while identifying accounts at risk of delinquency is particularly difficult, as even borrowers with historically strong credit profiles may begin missing payments. During economic downturns, these challenges become more pronounced as delinquency and potential losses increase, requiring credit unions to identify opportunities for proactive intervention and recovery before loans progress to default or charge-off.One potentially overlooked segment is members with relatively high credit scores who exhibit emerging delinquency. Rather than assuming that these borrowers represent lower recovery risk, this study investigates whether their delinquency patterns, outstanding loan balances, and account characteristics may identify them as a priority segment for early intervention and potential recovery. This approach aims to support more targeted loan-management and recovery strategies during periods of economic uncertainty.

These challenges demand a more sophisticated, forward-looking analytical approach. This research develops and validates an integrated credit risk framework - presented as a single-institution case study - combining three complementary stochastic methods: Markov Chain migration modeling, Monte Carlo loss simulation, and Bayesian behavioral anomaly detection. Unlike models relying solely on historical data, these techniques incorporate dynamic, probabilistic reasoning suited to evolving risk conditions.

Markov Chain modeling forecasts delinquency state transitions, capturing how accounts migrate across risk stages over time. Monte Carlo simulation generates full loss distributions under adverse economic scenarios, enabling portfolio-level loss estimation and capital reserve evaluation. The Bayesian component serves a focused operational purpose: detecting accounts where external credit bureau scores and internal payment behavior diverge - identifying members whose actual repayment risk within the credit union is systematically understated by traditional scoring alone.

Applied to 93,971 active loan accounts observed in 13 month-end snapshots from August 2025 to August 2026 (926,155 state transitions), the Markov Chain achieved an out-of-sample AUC of 0.86 for default and 0.92 for serious delinquency. Monte Carlo simulation on 51,966 accounts ($1.94 billion exposure) estimates 12-month expected losses of $1.86 million, $13.28 million, and $48.84 million under baseline,

adverse, and severely adverse scenarios, reconciling within 0.6% of the analytical expectation. The Bayesian component identified 4,621 accounts representing $91.1 million in exposure whose credit scores (average 762) qualified them as safe under standard decisioning, yet which were 3.8 times more likely to be past due than accounts with normal payment behavior. Kullback–Leibler drift monitoring confirmed that the calibrated transition matrix remains representative of the current portfolio.

1. Introduction

1.1 Credit Unions and Risk Management Context

Credit unions are member-owned, nonprofit financial cooperatives that provide banking services — including savings and loans — with a focus on community, financial inclusion, and member well-being. Unlike traditional banks, credit unions typically offer higher interest rates on deposits and lower fees and interest rates on loans. Their primary purpose is to provide reasonably priced financial access, improve financial literacy, strengthen local economies through small business lending, and serve underserved populations. They also foster economic resilience during events such as pandemics, climate events, and economic downturns, while providing customized financial guidance to members.

In the United States, credit unions are regulated by the National Credit Union Administration (NCUA). As credit unions grow larger and more complex, NCUA regulations evolve to maintain the strength and stability of the credit union system. The Current Expected Credit Loss (CECL) standard (FASB ASU 2016-13) requires credit unions to estimate expected credit losses over the life of their loans at origination or acquisition. Despite their important role, credit unions face significant data and resource constraints compared to large banks. While large banks can implement modern data storage and advanced analytics platforms, many credit unions continue to rely on legacy monolithic core systems, often requiring third-party assistance for ETL processes to migrate data to cloud platforms.

1.2 The Credit Risk Challenge

When a loan is originated, repayment is expected across various product categories including auto loans (direct and indirect), personal loans, credit cards, HELOCs, mortgages, and other secured and unsecured products. Under CECL guidelines, credit unions typically employ point-estimate techniques based on management judgment and GAAP-compliant models. Significant post-origination challenges emerge when executives must address:

- How will borrower accounts migrate across delinquency stages in future months?
- What is the probability that certain accounts will default under normal and stressed economic environments?
- How much capital should be reserved to absorb losses under recession-level unemployment?
- Which individual accounts display behavioral patterns that contradict their credit scores and require early intervention?
- Is the calibrated model still representative of the portfolio, or has the portfolio drifted enough to require recalibration?

Most credit unions depend on simplified scorecards, static credit scores, traditional delinquency reports, or vendor black-box models that provide limited interpretability and no mechanism for dynamically updating borrower risk as behavior changes.

1.3 Research Contribution

Prior research has explored various aspects of credit risk modeling. Markov Chain models have demonstrated that loan performance follows state-dependent transition patterns, while Monte Carlo simulation has been applied to portfolio-level loss estimation. Bayesian methods have shown promise for incorporating behavioral signals into risk assessments. However, no prior framework integrates all three methods into a cohesive post-origination system with behavioral anomaly detection at the account level.

The three models in this framework are designed sequentially and implemented as a single production pipeline in Snowflake (Snowpark Python), with each model building upon the outputs of the previous one through a shared data foundation. The Markov Chain model establishes structural transition probabilities from a point-in-time panel of 12 months of month-end delinquency states. The Monte Carlo simulation reads the same transition matrix, applies scenario-specific stress, and simulates account-level state paths to generate portfolio loss distributions. The Bayesian behavioral detection model uses the Markov Chain's 12-month default probability as its prior and the Monte Carlo recovery parameters as its loss severity, updating the prior with 12 months of individual payment history. A fourth, supervisory component — Kullback–Leibler (KL) divergence monitoring — tests whether the portfolio has drifted far enough from the calibration window to require recalibration. The key contribution lies in integrating these methods into a cohesive, governed framework for post-origination risk management — a gap identified in existing literature.

2. Literature Review and Research Gap

2.1 Pre-Loan versus Post-Loan Risk Assessment

The literature on credit risk modeling has developed along two distinct trajectories: pre-origination assessment and post-origination portfolio management. Pre-origination models focus on predicting default probability at loan application, while post-origination models address ongoing portfolio monitoring and loss forecasting. Research by Chen et al. (2020) developed a Bayesian MCMC model with Hamiltonian Monte Carlo to predict pre-loan default using 255,000 loan records. Their model incorporated variables including credit score, income, loan amount, employment months, debt-to-income ratio, interest rate, education, loan purpose, and marital status. The study demonstrated the efficacy of Bayesian methods for origination decisions, achieving high predictive accuracy through sophisticated preprocessing including missing value imputation, categorical variable encoding, and numeric field normalization.

While valuable for underwriting decisions, this approach does not address the post-origination challenge. After loans are disbursed to thousands of customers, credit unions require tools to monitor portfolio health, identify deteriorating accounts, and estimate losses under stress scenarios. The integrated framework proposed in this research addresses this gap by enabling credit unions to understand post-loan

portfolio dynamics, identify risk at the account level, and estimate losses under varying economic conditions.

2.2 Portfolio-Level Risk Estimation

Several studies have employed Monte Carlo methods for portfolio-level risk estimation. Lee and Kim (2019) proposed a simplified approach to calculating credit risk in lending portfolios, demonstrating how Monte Carlo simulation can estimate portfolio risk. However, their method replaced full simulation with analytical approximation using a single systematic risk factor (economy). While computationally efficient and useful for capital estimation, this approach does not predict which specific borrowers will default or provide account-level loss estimates essential for targeted interventions.

2.3 Migration-Based Credit Risk Models

Markov Chain models have been extensively applied to credit risk migration analysis. Schuermann and Jafry (2003) examined examples of migration matrix models and their performance in credit risk analysis, highlighting both their utility and limitations. Similarly, Guzmán and Trujillo (2018) estimated credit quality transition matrices for commercial loan portfolios in the Colombian financial system, demonstrating the applicability of Markov methods in emerging markets.

Both studies discuss limitations of Markov Chain models, particularly their inability to incorporate economic and behavioral covariates without extension. This research extends migration-based modeling by integrating Monte Carlo simulation for loss distribution estimation and Bayesian methods for dynamic account-level risk assessment. Migration-based models are primarily designed for post-origination risk monitoring rather than loan approval decisions, making them complementary to the framework proposed herein.

2.4 Bayesian Credit Risk Applications

Bayesian methods have gained traction in credit risk modeling due to their ability to incorporate prior information and update beliefs as new data emerge. Karanikas and Theodossiou (2012) applied Bayesian approaches to modeling credit risks, estimating borrower-level default probability based on characteristics such as income, age, employment experience, loan amount, interest rate, credit rating, loan-to-income ratio, housing type, credit history, and loan intention using publicly available credit data. While demonstrating the power of Bayesian inference for individual risk assessment, this research does not address delinquency migration patterns, portfolio loss estimation, or capital adequacy under stress scenarios.

2.5 Comprehensive Risk Frameworks

Rodriguez and Torres (2021) quantified credit risk in the lending industry using a Monte Carlo simulation approach that addressed both pre-loan and post-loan risk estimation. Their framework used factors such as credit score, collateral, and debt-to-income ratio to understand borrower risk and set appropriate interest rates through risk-based pricing. While comprehensive in scope, this approach lacks the dynamic updating capability that Bayesian methods provide, relying instead on static risk classifications.

2.6 Computational Efficiency in Monte Carlo Methods

Monte Carlo methods for portfolio credit risk have also evolved in terms of computational efficiency. Brereton et al. (2014) focused on accelerating Monte Carlo simulation through variance reduction techniques such as importance sampling and cross-entropy methods for rare event estimation. However, less attention has been paid to continuously updating input parameters based on observed borrower behavior.

This research employs standard Monte Carlo methods with scenario-based stress testing across multiple economic scenarios. By modeling dependencies through macroeconomic scenarios (unemployment rates) that affect all borrowers simultaneously, the framework generates actionable portfolio- and account-level insights for portfolio management.

Recent studies have explored the use of Markov Chain Monte Carlo (MCMC) methods combined with Bayesian inference for credit risk modeling. Moolchandani (2023) demonstrated that MCMC techniques improve the estimation of probability of default by sampling from posterior parameter distributions and capturing dependencies among financial variables. The study highlights the advantages of Bayesian modeling over traditional statistical approaches such as logistic regression. However, the research primarily focuses on parameter estimation for default prediction rather than post-origination portfolio monitoring. In contrast, the present study develops a comprehensive framework integrating Markov Chain migration analysis, Bayesian updating, and Monte Carlo simulation to evaluate borrower behavior and portfolio-level credit risk in credit unions.

2.7 Model Monitoring and Distributional Drift

Transition matrices are estimated on a finite window and assume stationarity. Information-theoretic divergence measures — the Kullback–Leibler divergence (Kullback & Leibler, 1951) and its symmetric, bounded counterpart, the Jensen–Shannon divergence (Lin, 1991) — provide a direct test of whether the distribution a model was calibrated on still resembles the distribution it is applied to. Such drift monitoring is rarely paired with migration models in the credit union literature.

2.8 The Research Gap

The literature reveals that while individual methods have been applied to credit risk, an integrated framework combining Markov Chain migration modeling, Monte Carlo loss simulation, Bayesian behavioral updating, and formal drift monitoring for post-origination portfolio management has not been developed. Existing approaches treat these components separately or omit critical elements such as dynamic updating, account-level granularity, or recalibration triggers. This research addresses this gap by developing and validating an integrated framework specifically designed for credit union portfolio management.

3. Data Description

3.1 Participating Institution

The study uses loan portfolio data from a credit union operating in the United States, hereafter referred to as Credit Union A, to preserve institutional confidentiality. The dataset was obtained through the institution's cloud analytics layer (Snowflake), which receives daily loads from the core system. All member-level identifiers were removed prior to reporting. The credit union operates as a member-owned financial cooperative and often maintains close relationships with borrowers, meaning borrower behavior may differ from patterns observed in large banking institutions.

The source loan account table is maintained as a change history: each row carries a validity period (row begin and end dates), and a new row is written only when an account attribute changes. A point-in-time monthly panel was therefore constructed by taking, for every account and each of the 13 month-ends from August 31, 2025 to August 31, 2026, the row valid on that date. The resulting panel contains 179,865 accounts. After excluding closed accounts (retaining charged-off accounts), the analytical panel contains 1,020,130 account-month observations for 93,971 accounts, an average of 10.9 months per account. The reference date for all models is August 31, 2026.

3.2 Analytical Samples and Sequential Filtering Pipeline

Separate analytical samples were constructed for each model. The differences in sample sizes are intentional — each model applies a filter appropriate to its analytical purpose, forming a sequential pipeline shown in Table 1.

Step	Sample	Size	Filter Applied	Purpose
1	Point-in-Time Universe	179,865 accounts	Row valid at each of 13 month-ends (Aug 2025 – Aug 2026)	Portfolio context and history
2	Markov Chain Sample	93,971 accounts; 926,155 transitions	Closed accounts excluded (charge-offs retained); consecutive month-end pairs; no exits from Default	Transition probability estimation
3	Monte Carlo Active Sample	51,966 accounts ($1.94B)	Latest state per account; balance > $0; excludes Default	Loss simulation requires positive exposure
4	Bayesian Scoring Sample	51,967 accounts ($1.94B)	Latest account row; balance > $0; excludes Default	Account-level default probability and expected loss
5	Behavioral Evidence Subset	29,070 accounts ($570.1M)	At least one qualifying transaction record in the 12-month window	Behavioral anomaly flags and prioritization

Table 1: Sequential Filtering Pipeline — Sample Sizes and Rationale. The Monte Carlo and Bayesian samples differ by one account because they derive the latest state from the monthly panel and the latest source row, respectively.

3.3 Data Sources and Key Variables

The analytical dataset was constructed by integrating five operational tables from Credit Union A's core system:

- **Loan Account:** Account_ID, Member_ID, Account Status Code (Open, Closed, Charge_Off), Snapshot Date, Row Begin/End Date, Account Open Date, Account Closed Date, Delinquency Begin Date, Delinquency End Date, Charge-off Date, Product Code, Original Balance Amount.

- **Loan Transaction:** Account_ID, Transaction Posting Date, Transaction Amount, Transaction Type, Transaction Sub-Type.
- **Account Balance:** Account_ID, Balance Date, Balance Amount.
- **Member Score:** Member_ID, Credit Score (numeric value), Snapshot Date.
- **Product:** Product Code, Is_Secured flag (secured, unsecured, or not applicable).

Variable	Definition	Source Table	Model Role
ACCOUNT_STATE	Five-state delinquency classification at each month-end	LOAN_ACCOUNT	Markov state; Bayesian prior lookup
CREDIT_SCORE	Latest bureau score (650 imputed if missing)	MEMBER_SCORE	Anomaly detection baseline
CURRENT_BALANCE	Latest balance (fallback: original balance)	ACCOUNT_BALANCE	Exposure at Default (EAD)
DAYS_PAST_DUE	Reference date − delinquency begin date (0 if cured)	LOAN_ACCOUNT	Delinquency multiplier
PAYMENT_CONSISTENCY	Share of months with a qualifying payment (12-month window, capped at 1)	LOAN_TRANSACTION	Behavioral likelihood evidence
IS_SECURED	Secured flag (Y / N / not applicable)	PRODUCT	LGD and recovery distribution

Table 2: Key Variables Used in the Analysis

Table Name	Fields Used	Purpose in Models
Loan Account	Account ID, Row Begin/End Date, Status, Closed Date, Charge-off Date, Delinquency Begin/End Date, Original Balance	Point-in-time state classification, transition tracking
Account Balance	Account ID, Balance Amount, Balance Date	Current exposure (EAD)
Product	Product Code, Is_Secured	Secured/unsecured segmentation, LGD assignment
Loan Transaction	Account ID, Transaction Amount, Posting Date, Type, Sub-Type	Payment consistency, evidence weight, behavioral flags
Member Score	Member ID, Score Numeric Value, Snapshot Date	Credit score for anomaly detection

Table 3: Data Sources and Field Usage

3.4 Sample Selection Criteria

Accounts were included in the Markov panel if a loan account row was valid at the month-end and the account had been opened on or before that date. Accounts closed on or before the month-end were excluded, except where the account had been charged off, so that the absorbing Default state is retained. For the Monte Carlo and Bayesian samples, accounts were further required to have a positive current balance and a non-default state at the reference date. Qualifying payment transactions are defined as Transaction Type = CREDIT, Sub-Type = PAYMENT, and amount greater than zero.

3.5 Portfolio Characteristics and Descriptive Statistics

At the August 31, 2026 reference snapshot, the 80,560 accounts in the Markov panel were distributed as 62.52% Current, 11.41% DPD_30, 0.75% DPD_60, 6.17% DPD_90+, and 19.15% Default (accumulated charge-offs). The Bayesian scoring sample of 51,967 non-default accounts with positive balance exhibits the following characteristics:

- Total exposure: $1,937,274,210
- Composition: 66.7% secured (34,648 accounts, $1,845.9M) and 33.3% unsecured or not applicable (17,319 accounts, $91.3M); conventional mortgages account for 4,406 accounts and $639.0M
- Delinquency distribution: 84.5% Current (43,924), 14.1% DPD_30 (7,310), 0.8% DPD_60 (431), 0.6% DPD_90+ (302)
- Credit score: median 733; 33,495 accounts (64.5%) at or above 700; 5,840 accounts carry the imputed score of 650
- Transaction coverage: 29,070 accounts ($570.1M) have transaction records in the 12-month window; 22,897 accounts ($1.37B) have none (Section 6.3)
- Average payment consistency among accounts with transaction data: 91.6%

3.6 Delinquency State Classification

Loan accounts are classified into five discrete delinquency states consistent with standard credit risk literature (Jarrow, Lando & Turnbull, 1997). The same classification — with charge-off evaluated before cure — is applied in the Markov, Monte Carlo, Bayesian, and KL monitoring code, so every account is classified identically across models.

State	Rule (evaluated in order)
DEFAULT	Status = Charge_Off, or a valid charge-off date on or before the month-end (absorbing)
CURRENT	No valid delinquency begin date, days past due ≤ 0, or delinquency end date on or before the month-end (cured)
DPD_30	1–29 days past due
DPD_60	30–59 days past due
DPD_90+	60 or more days past due

Table 4: Delinquency State Definitions. The DPD_90+ label is retained from the production schema and covers all accounts 60 or more days past due.

4. Methodology

This section presents the mathematical and theoretical formulation of the three models, the drift monitor, and the validation procedures. All data description and variable definitions are consolidated in Section 3. All models share the August 31, 2026 reference date and identical state classification logic.

4.1 Model Integration Framework

The three models are designed as a sequential, hierarchically dependent system. Figure 1 illustrates the data flow. Each layer reads its inputs from Snowflake tables written by the previous layer, so each stage can be re-run independently and every run is time-stamped.

LAYER 1 **Markov Chain** Point-in-time month-end panel State pairs (s_t, s_{t+1}) MLE: P_ij = N_ij / N_i Outputs: • 5×5 transition matrix P • 12-month PD by state (P^{12}) • 12-month state forecast	→	**LAYER 2** **Monte Carlo** Reads P from Snowflake Scenario-stressed P Beta recovery (secured / unsecured) 2,000 paths × 12 months Outputs: • Expected loss, VaR 95/99 • Expected shortfall, defaults • 3 stress scenarios	→	**LAYER 3** **Bayesian Detection** Prior PD = P^{12} from Layer 1 LGD = Beta mean from Layer 2 12-month payment history Evidence-weighted update Outputs: • Final PD, EL per account • Anomaly flags + priority tier
INPUT: 93,971 accounts *926,155 transitions* *Aug 2025 – Aug 2026*		***INPUT: 51,966 accounts*** *$1.94B exposure* *Balance > $0, non-default*		***INPUT: 51,967 accounts scored*** *29,070 with transaction data* *Aug 31, 2026 reference*

Figure 1: Sequential Model Integration Framework — Data Flow Across Three Analytical Layers. All layers share the August 31, 2026 reference date and identical state classification logic.

4.2 Markov Chain Model: Delinquency Migration

4.2.1 Theoretical Foundation

A Markov chain is a discrete-time stochastic process satisfying the Markov property — the future state depends only on the current state, not on the history by which it was reached. Formally:

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = P(X_{t+1} = j \mid X_t = i) = P_{ij}$$

The state space is S = {Current, DPD_30, DPD_60, DPD_90+, Default}, where Default is an absorbing state (Jarrow, Lando & Turnbull, 1997). Transitions out of Default are removed before estimation.

4.2.2 Transition Probability Estimation

Transition probabilities are estimated using maximum likelihood estimation (MLE), standard for discrete-time Markov chains (Anderson & Goodman, 1957), over all consecutive month-end pairs in the 12 months ending at the reference date:

$$P_{ij} = N_{ij} / N_i$$

where N_ij is the number of observed transitions from state i to state j and N_i is the total number of transitions from state i. All 25 state pairs are generated explicitly, so unobserved transitions are recorded as zero. Confidence intervals are computed using the Wilson score interval (Wilson, 1927), where z = 1.96 for 95% confidence:

$$CI = \left(\hat{p} + z^2/2n \pm z \cdot \sqrt{\left(\hat{p}(1-\hat{p})/n + z^2/4n^2 \right)} \right) / (1 + z^2/n)$$

4.2.3 Forecasting Algorithm

$$\pi(t+n) = \pi(t) \cdot P^n$$

Multi-step forecasts are computed by matrix power from the reference-date distribution. The Default column of P^{12} gives the 12-month cumulative probability of default (PD) by starting state, which is used as the Bayesian prior. The convergence rate towards the long-run distribution is governed by the second-largest eigenvalue of P.

4.3 Monte Carlo Portfolio Loss Simulation

4.3.1 Theoretical Foundation and Loss Function

Monte Carlo simulation generates full distributions of portfolio losses by conducting repetitive stochastic experiments (Glasserman, 2003). Rather than sampling a single default event, the implementation simulates each account's monthly state path through the (stressed) transition matrix over a 12-month horizon:

$$\text{Loss}_n = \Sigma_i \ \text{EAD}_i \times 1\{\text{account i in Default at month 12 on path n}\} \times (1 - \text{RR}_{i,n})$$

where EAD is the current balance and RR the recovery rate. Each of the 51,966 active accounts is simulated for 2,000 independent paths per scenario. State updates are computed against a start-of-month snapshot so that each account makes exactly one transition per month. Accounts are processed in memory-bounded batches.

4.3.2 Recovery Rate Modeling via Beta Distribution

Parameter	Secured Loans	Unsecured Loans	Interpretation
Alpha	7	2	Prior weight on recovery
Beta	3	8	Prior weight on loss
Mean Recovery Rate	70%	20%	$\alpha / (\alpha + \beta)$
Implied LGD	30%	80%	1 − mean recovery; also used as the Bayesian LGD

Table 5: Beta Distribution Parameters for Recovery Rate Modeling. Accounts with a not-applicable secured flag are treated as unsecured.

4.3.3 Economic Scenario Design and Matrix Stress

Scenario	Unemployment Rate	Historical Analog	PD Multiplier
Baseline	4.3%	Normal economy	1.0×
Adverse	7.0%	Mild recession (2001)	2.0×
Severely Adverse	10.0%	Deep recession (2008–2009)	3.5×

Table 6: Economic Stress Scenarios

Stress is applied to the transition matrix itself. For each origin state, every transition to a worse state is multiplied by the scenario multiplier (capped at 1). If the stressed worsening probability mass is below 1, the remaining probability is redistributed across non-worsening transitions in proportion to their baseline values; otherwise the worsening transitions are renormalized to sum to 1. Rows remain stochastic and

Default remains absorbing. Recovery-rate parameters are held at their base values in all scenarios, so scenario differences reflect default frequency only.

4.3.4 Risk Metrics

$$EL = (1/N) \cdot \Sigma\ Loss_n \qquad VaR_\alpha = \alpha\text{-percentile of } \{Loss_n\} \qquad ES_\alpha = E[\ Loss \mid Loss \geq VaR_\alpha\ ]$$

4.4 Bayesian Behavioral Anomaly Detection

4.4.1 Purpose: Detecting Credit Score Divergence

The Bayesian component serves a specific, bounded operational purpose: detecting accounts where external credit bureau scores and observed internal payment behavior convey conflicting risk signals. Bureau scores aggregate payment behavior across all of a member's creditors. A member can maintain a high bureau score while simultaneously exhibiting weak payment engagement specifically with this credit union - a risk invisible to score-based decisioning but detectable through the credit union's own transaction records. Using Bayes' theorem:

$$P(\text{Default} \mid \text{Payment Behavior}) \propto P(\text{Payment Behavior} \mid \text{Default}) \times P(\text{Default})$$

The prior P(Default) is derived from the account's current delinquency state using the Markov Chain model. The likelihood is constructed from 12 months of internal transaction history. The posterior combines structural delinquency risk with behavioral signals, serving two purposes: (1) Detection — identify accounts where bureau score and internal behavior diverge; and (2) Prioritization — rank urgency among detected accounts.

4.4.2 Prior Probability of Default

Prior PDs are read directly from the Markov Chain as the 12-month cumulative default probability (Default column of P^{12}), matching the Monte Carlo 12-month horizon and ensuring cross-model consistency:

Delinquency State	Prior PD (12-month)	Bayesian Update Rule	Max Posterior Cap
Current	0.198%	Consistency-adjusted × delinquency multiplier	85%
DPD_30 (1–29 days)	1.110%	Consistency-adjusted × delinquency multiplier	85%
DPD_60 (30–59 days)	6.282%	Consistency-adjusted × delinquency multiplier	85%
DPD_90+ (60+ days)	10.239%	Consistency-adjusted × delinquency multiplier	85%

Table 7: Prior PD by Delinquency State — 12-Month Cumulative PD from the Markov Chain (P^{12})

4.4.3 Behavioral Evidence and Posterior PD Calculation

1. **Payment Consistency:** months with at least one qualifying payment divided by min(12, months on book), capped at 1.
2. **Evidence Weight (ω):** min(payment count / 12, 1).

3. **Payment Adequacy:** min(total paid / (10% × current balance), 1).
4. **Delinquency Multiplier (DM):** 1.0 (0 days), 1.5 (1–30), 2.0 (31–60), 2.5 (61–90), 3.0 (over 90).

Condition (first match)	Posterior PD
Payment adequacy < 0.5	Prior × 2.0 × min(DM, 3.0)
Evidence weight < 0.1	Prior × min(DM, 2.0)
Consistency ≥ 0.95	Prior × 0.8 × min(DM, 1.5)
Consistency ≥ 0.85	Prior × 0.9 × min(DM, 2.0)
Consistency ≥ 0.70	Prior × 1.0 × min(DM, 2.5)
Consistency ≥ 0.50	Prior × 1.2 × min(DM, 3.0)
Consistency > 0	Prior × 1.5 × min(DM, 4.0)
No payments	Prior × 2.0 × min(DM, 5.0)

Table 8: Posterior PD Update Rules (posterior capped at 0.85)

Final PD = (1 − ω) × Prior PD + ω × Posterior PD **Expected Loss = Balance × Final PD × LGD**

Accounts are assigned to High Risk (Final PD ≥ 20%), Medium Risk (≥ 10%), or Low Risk. Accounts with no transaction records carry an evidence weight of zero and therefore retain the Markov prior.

4.4.4 Behavioral Anomaly Classification and Prioritization

Flag	Definition	Recommended Action
Insufficient Transaction Data	No qualifying payments AND 3 or fewer transactions of any type in 12 months	Excluded from flagging; retains Markov prior
High Score, Poor Behavior	Score ≥ 700 AND payment consistency < 85% AND not in default	Monitor closely; escalate before formal delinquency
Low Score, Excellent Behavior	Score < 650 AND payment consistency ≥ 95% AND Current	Re-evaluate for new lending
Declining Payment Pattern	Consistency < 70% AND active months < 8.4 of 12	Hardship program outreach

Table 9: Behavioral Anomaly Flag Definitions (evaluated in order) and Recommended Actions

The Insufficient Transaction Data rule separates accounts with no evidence from accounts that are not paying: an account with no records gives the model nothing to evaluate and is not labeled as poor behavior. Flagged accounts are ranked with a priority score combining three percentile components and a flag-type weight (1.0 High Score/Poor Behavior, 0.8 Declining, 0.3 Low Score/Excellent):

Priority = 100 × w_type × (0.45 · rank(Final PD) + 0.35 · rank(Balance) + 0.20 · min(DPD,120)/120)

Scores above 66 are tiered P1 – Immediate Review, 33–66 P2 – Review This Cycle, and below 33 P3 – Monitor.

4.5 Model Drift Monitoring: KL Divergence

Drift is measured with the Kullback–Leibler divergence of the current distribution P from the reference distribution Q, and with the Jensen–Shannon divergence (bounded by ln 2 ≈ 0.693):

$$\mathbf{KL(P\|Q) = \Sigma_s\ P(s) \cdot \ln(\ P(s) / Q(s)\ )} \qquad \mathbf{JS = \tfrac{1}{2}\ KL(P\|M) + \tfrac{1}{2}\ KL(Q\|M),\ \ M = \tfrac{1}{2}(P + Q)}$$

Four signals are computed from the monthly panel, anchored to the reference date: (1) state-distribution drift at 9, 6 and 3 months ago and today against a September 2025 baseline; (2) per-state contribution to the current KL; (3) month-over-month KL, with spikes defined as values above the mean plus two standard deviations; and (4) row-wise KL between consecutive quarterly transition matrices. Values below 0.05 are Stable, 0.05–0.10 Watch, and above 0.10 Review. Recalibration is recommended when two or more of the three headline signals are in Review. September 2025 is used as the baseline because the August 2025 month-end is only partially loaded (59,466 accounts versus approximately 80,000 thereafter).

4.6 Model Validation Procedures

- **Markov Chain:** Wilson confidence intervals; a first-order Markov property test (chi-square test of whether the next state depends on the previous state, conditional on the current state); temporal stability (Frobenius norm between half-year matrices and quarterly KL); an out-of-sample test in which the matrix is estimated on September 2025 – March 2026 transitions and used to forecast the March 2026 cohort to August 2026 (distribution MAE and AUC); an in-sample 11-month back-test; and lookback-window sensitivity.
- **Monte Carlo:** convergence of the running mean and standard error of the mean; reconciliation of simulated expected defaults and expected loss against the analytical expectation Σ EAD × P^{12} × LGD.
- **Bayesian:** cross-model consistency with the Monte Carlo baseline; monotonicity of Final PD across delinquency states; comparison of the share of accounts past due at the reference date across behavioral flags.

5. Results

5.1 Markov Chain Results: Delinquency Migration

The Markov Chain was estimated from 926,155 observed month-end transitions between August 2025 and August 2026. Table 10 reports the full matrix and Table 11 the reliability of key probabilities.

From \ To	Current	DPD_30	DPD_60	DPD_90+	Default	N
Current	0.9568	0.0426	0.0006	0.0000	0.0000	599,022
DPD_30	0.1719	0.7736	0.0539	0.0005	0.0001	100,194
DPD_60	0.1434	0.1846	0.1920	0.4760	0.0040	7,210
DPD_90+	0.0268	0.0045	0.0041	0.9538	0.0108	54,399
Default	0.0000	0.0000	0.0000	0.0000	1.0000	165,330

Table 10: Estimated One-Month Transition Probability Matrix (5-state, row-stochastic; N = transitions out of each state)

Transition	Estimate	95% CI (Wilson)	Observed Count	Reliability
Current → Current	95.68%	[95.62%, 95.73%]	573,115	Reliable
Current → DPD_30	4.26%	[4.21%, 4.32%]	25,548	Reliable
DPD_30 → Current (cure)	17.19%	[16.96%, 17.43%]	17,227	Reliable
DPD_30 → DPD_30	77.36%	[77.10%, 77.61%]	77,506	Reliable
DPD_60 → DPD_90+	47.60%	[46.45%, 48.75%]	3,432	Reliable
DPD_60 → Default	0.40%	[0.28%, 0.58%]	29	Directional — sparse
DPD_90+ → DPD_90+	95.38%	[95.20%, 95.55%]	51,884	Reliable
DPD_90+ → Default	1.08%	[0.99%, 1.17%]	585	Reliable

Table 11: Key Transition Probabilities with 95% Wilson Confidence Intervals

Current accounts are highly persistent (95.68%), rolling to DPD_30 at 4.26% per month. DPD_30 accounts cure at 17.19% per month and roll forward at 5.39%. DPD_60 is the least stable state: 47.60% of its transitions move to DPD_90+ and 14.34% cure. DPD_90+ is highly persistent (95.38%) and is the main gateway to Default (1.08% per month; 585 of the 625 observed transitions into Default). With the expanded point-in-time panel, every confidence interval is narrow except the sparse DPD_60 → Default cell.

Month	Current	DPD_30	DPD_60	DPD_90+	Default
Aug 2026 (actual)	62.52%	11.41%	0.75%	6.17%	19.15%
Month 1	62.05%	11.66%	0.82%	6.25%	19.22%
Month 2	61.66%	11.85%	0.85%	6.36%	19.29%
Month 3	61.32%	11.98%	0.86%	6.47%	19.36%
Month 6	60.53%	12.17%	0.89%	6.83%	19.59%
Month 12	59.43%	12.16%	0.89%	7.44%	20.08%

Table 12: Portfolio State Forecast (Baseline Conditions, 4.3% Unemployment; Default includes accumulated charge-offs)

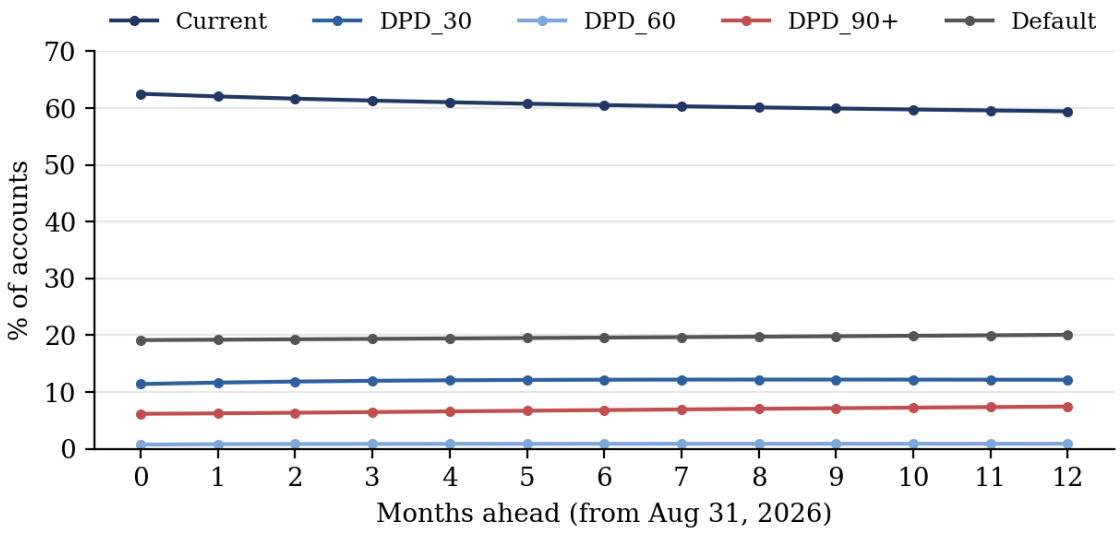


Figure 2: Projected State Distribution over 12 Months

Under baseline conditions the Current share is projected to decline gradually from 62.52% to 59.43% over 12 months, with DPD_90+ rising from 6.17% to 7.44% and the Default share from 19.15% to 20.08%. The 12-month cumulative probability of default is 0.198% for Current, 1.110% for DPD_30, 6.282% for DPD_60, and 10.239% for DPD_90+, increasing monotonically with delinquency severity. The second-largest eigenvalue of P is 0.9987, indicating slow convergence to the long-run distribution.

5.2 Monte Carlo Results: Loss Distribution Under Stress Scenarios

Transition	Baseline	Adverse (2.0×)	Severely Adverse (3.5×)
Current → DPD_30	0.0426	0.0853	0.1493
DPD_30 → DPD_60	0.0539	0.1078	0.1886
DPD_30 → Current	0.1719	0.1620	0.1472
DPD_60 → DPD_90+	0.4760	0.9520	0.9861
DPD_90+ → Default	0.0108	0.0215	0.0376

Table 13: Selected Stressed Transition Probabilities

Metric	Baseline (4.3%)	Adverse (7.0%)	Severely Adverse (10.0%)
Expected Loss (mean)	$1.86M	$13.28M	$48.84M
EL % of Exposure ($1.94B)	0.10%	0.69%	2.52%
Median Loss	$1.83M	$13.21M	$48.70M
Standard Deviation	$0.36M	$0.99M	$1.92M
VaR (95th)	$2.39M	$14.92M	$52.27M
VaR (99th)	$2.83M	$16.65M	$54.16M

Metric	Baseline (4.3%)	Adverse (7.0%)	Severely Adverse (10.0%)
Expected Shortfall (95%)	$2.75M	$15.82M	$53.46M
Expected Shortfall (99%)	$3.66M	$17.64M	$54.74M
Expected Defaults (of 51,966)	226	1,371	4,553
EL Multiple of Baseline	1.0×	7.1×	26.3×

Table 14: Monte Carlo Simulation Results by Economic Scenario (51,966 accounts, 2,000 paths, 12-month horizon)

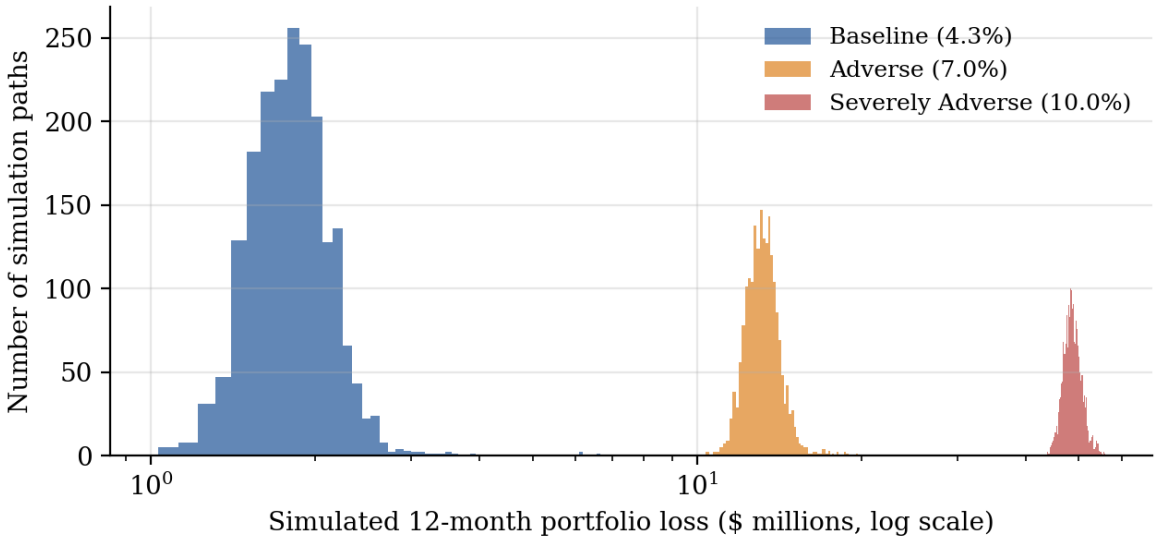


Figure 3: Distribution of Simulated 12-Month Portfolio Loss by Scenario

Baseline expected loss is $1.86 million (0.10% of exposure), with a 99th-percentile loss of $2.83 million. Losses scale strongly super-linearly with the PD multiplier: a 2.0× multiplier raises expected loss 7.1-fold and a 3.5× multiplier 26.3-fold, to $48.84 million (2.52% of exposure). This reflects compounding: the multiplier applies to every worsening transition over 12 monthly steps, and under stress almost every DPD_60 account rolls into DPD_90+, the main gateway to Default. Because recovery parameters are not stressed, these estimates are conservative in the sense of understating downturn severity.

5.3 Bayesian Results: Account-Level Risk Assessment

Metric	Value	Notes
Total Accounts Scored	51,967	Balance > $0, non-default at the reference date
Total Portfolio Exposure	$1,937,274,210	Sum of current balances
Total Expected Loss	$2,105,939	EL = Balance × Final PD × LGD
Portfolio Loss Rate	0.11%	Within 13% of the Monte Carlo baseline
Average Prior / Posterior / Final PD	0.43% / 1.01% / 0.52%	Exposure-weighted Final PD 0.32%

Metric	Value	Notes
Accounts with Transaction Data	29,070 ($570.1M)	Average payment consistency 91.6%
Risk Category	143 High / 387 Medium / 51,437 Low	High Risk: Final PD ≥ 20%

Table 15: Bayesian Model Portfolio Summary

State	Count	Exposure	Expected Loss	Prior PD	Avg. Final PD
Current	43,924	$1,858.9M	$1,144,176	0.20%	0.19%
DPD_30	7,310	$67.7M	$412,703	1.11%	1.25%
DPD_60	431	$4.3M	$169,151	6.28%	9.71%
DPD_90+	302	$6.3M	$379,909	10.24%	18.80%

Table 16: Risk Distribution by Delinquency State

Final PD rises monotonically with delinquency severity. For Current accounts the behavioral update slightly lowers the prior on average (0.19% versus 0.20%), reflecting strong payment consistency, while for delinquent accounts it raises the prior substantially (DPD_90+: 18.80% versus 10.24%), reflecting weak or missing payments. Secured loans carry an average Final PD of 0.42% and unsecured loans 0.72%.

5.4 Behavioral Anomaly Detection Results

Flag	Accounts	Exposure	Avg. Score	Avg. Final PD	Past Due	DPD_60+
High Score, Poor Behavior	4,621	$91.1M	762	1.19%	22.3%	4.1%
Declining Payment Pattern	956	$11.4M	510	2.97%	37.0%	13.1%
Low Score, Excellent Behavior	1,374	$20.2M	365	0.17%	0.0%	0.0%
Normal (comparison group)	22,119	$447.4M	734	0.33%	5.9%	0.7%
Insufficient Transaction Data	22,897	$1,367.2M	674	0.49%	n/a	n/a

Table 17: Behavioral Anomaly Detection Summary — 51,967 Accounts Scored. Past-due shares are measured at the reference date.

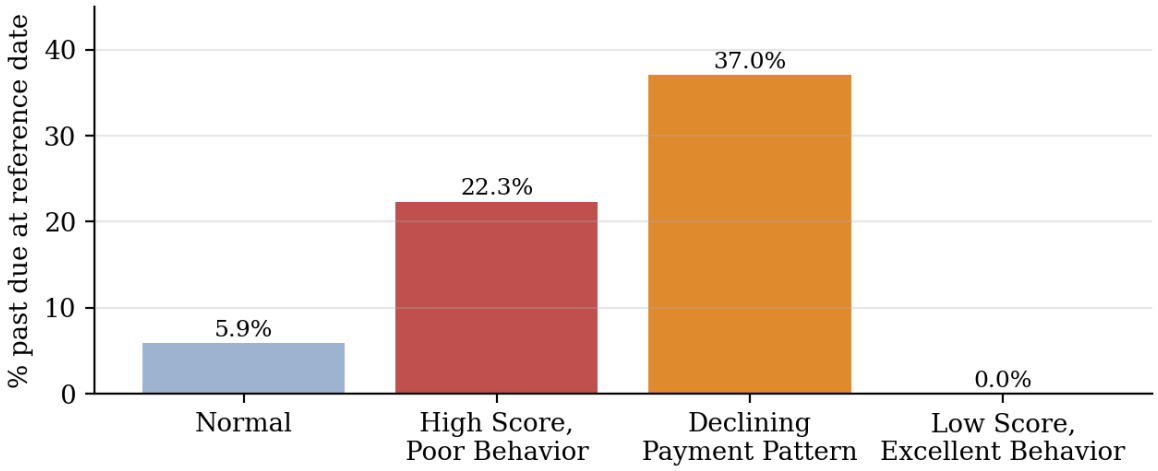


Figure 4: Share of Accounts Past Due at the Reference Date, by Behavioral Flag

The most strategically significant finding is the High Score, Poor Behavior cohort: 4,621 accounts representing $91.1 million in exposure, with an average credit score of 762. These members qualify as low-risk under standard score-based decisioning, yet 22.3% are past due — 3.8 times the rate among accounts with normal payment behavior (5.9%) — and 4.1% are 60 or more days past due, 6.1 times the normal rate (0.7%). Because these accounts sit above a 700 threshold, a rule-based system that uses credit score as the primary filter would classify every one of them as safe; the Bayesian framework's contribution is identifying risk that is invisible by design to score-based approaches.

The Declining Payment Pattern segment (956 accounts) carries the highest average Final PD (2.97%) and past-due rate (37.0%). The Low Score, Excellent Behavior segment (1,374 accounts, $20.2 million) has no past-due accounts and a Final PD of 0.17% despite an average score of 365, identifying members likely excluded by traditional underwriting who represent a lending opportunity.Accounts without transaction records (22,897) are not flagged and retain the Markov prior.

Priority Tier	Accounts	Exposure	Expected Loss	Avg. Final PD	Past Due
P1 – Immediate Review	470	$20.5M	$315,312	7.35%	88.7%
P2 – Review This Cycle	3,672	$76.2M	$155,133	1.25%	25.7%
P3 – Monitor	2,809	$26.1M	$16,310	0.20%	0.9%

Table 18: Anomaly Prioritization Tiers (6,951 flagged accounts)

The priority tiers concentrate risk effectively: the 470 P1 accounts (0.9% of the scored portfolio) hold $315,312 of expected loss — more than P2 and P3 combined — and 88.7% are already past due. Of the P1 accounts, 448 are High Score, Poor Behavior.

5.5 Portfolio Stability: KL Divergence Monitoring

Snapshot	Accounts	KL vs. Baseline	JS vs. Baseline	Status
9 months ago (Nov 2025)	80,263	0.00001	0.000003	Stable

Snapshot	Accounts	KL vs. Baseline	JS vs. Baseline	Status
6 months ago (Feb 2026)	79,831	0.03707	0.01045	Stable
3 months ago (May 2026)	80,094	0.00048	0.00012	Stable
Current (Aug 2026)	80,560	0.00061	0.00015	Stable

Table 19: State-Distribution Drift versus the September 2025 Baseline (80,006 accounts)

State	Baseline (Sep 2025)	Current (Aug 2026)	Change	KL Contribution
Current	62.31%	62.52%	+0.21 pp	+0.0021
DPD_30	12.06%	11.41%	−0.65 pp	−0.0063
DPD_60	0.78%	0.75%	−0.03 pp	−0.0003
DPD_90+	6.65%	6.17%	−0.48 pp	−0.0046
Default	18.20%	19.15%	+0.95 pp	+0.0097
Total KL				**0.0006**

Table 20: Per-State KL Contribution (Current vs. Baseline). Default is the largest single contributor.

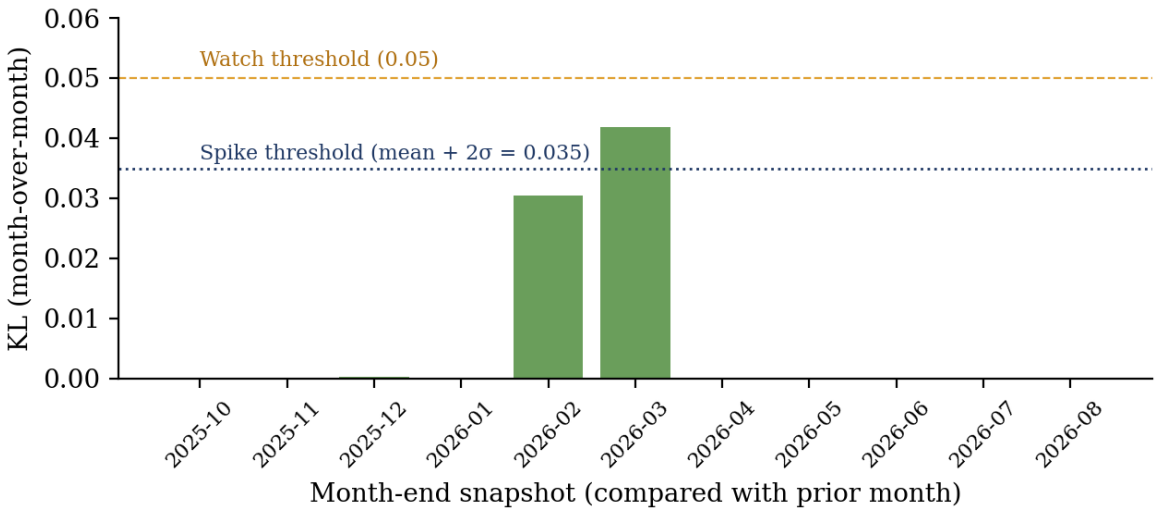


Figure 5: Month-over-Month KL Divergence, September 2025 – August 2026

All snapshots are Stable against the baseline (KL ≤ 0.037). Month-over-month KL averaged 0.0067; the only elevated values are January → February 2026 (0.030) and February → March 2026 (0.042, above the spike threshold of 0.035). These are a calendar effect rather than a shock: at the 28-day February month-end fewer accounts fall in the 1–29 days past due bucket (4.40% versus about 12%), and the share returns to normal in March. Quarterly transition matrices are also stable (average row KL 0.029, 0.023, and 0.003 for Q1→Q2, Q2→Q3 and Q3→Q4); the only row above 0.05 is DPD_30 in the two comparisons that include February.

Signal	KL Value	Status
Distribution drift (vs. Sep 2025 baseline)	0.0006	Stable
Latest month-over-month (Jul → Aug 2026)	0.00005	Stable
Transition matrix drift (latest quarter)	0.0029	Stable
Decision		**No action required**

Table 21: Recalibration Decision Summary

5.6 Model Validation Results

5.6.1 Markov Chain Validation

Test	Result	Interpretation
Out-of-sample distribution forecast (train Sep 2025 – Mar 2026; forecast Mar → Aug 2026; 74,451 accounts)	MAE 0.57 pp across states	Accurate portfolio-level forecast
Out-of-sample default ranking (59,385 non-default accounts; 244 defaults)	AUC 0.859	Strong discrimination
Out-of-sample serious delinquency ranking (DPD_60+ or Default)	AUC 0.921	Strong discrimination
In-sample 11-month back-test (Sep 2025 → Aug 2026; 54,318 accounts)	Distribution MAE 0.18 pp; default AUC 0.760	Consistent with out-of-sample results
Temporal stability (Sep 2025 – Feb 2026 vs. Feb – Aug 2026)	Frobenius norm 0.146; quarterly KL ≤ 0.029	Stable; half-year difference partly reflects the February calendar effect
First-order Markov property (previous state vs. next state, given current)	Rejected ($p < 0.001$); Cramér's V 0.15–0.28	Moderate path dependence; see Section 6.3

Table 22: Markov Chain Validation Summary

Starting State (Mar 2026)	Accounts	Actual 5-Month Default Rate	Predicted
Current	45,986	0.06%	0.02%
DPD_30	8,322	0.67%	0.27%
DPD_60	540	12.04%	2.86%
DPD_90+	4,537	2.09%	4.86%

Table 23: Out-of-Sample Calibration by Starting State

Rank ordering is strong, but calibration is uneven: the model under-predicts default for Current, DPD_30 and especially DPD_60 accounts and over-predicts it for DPD_90+ accounts, where many long-delinquent accounts remain delinquent without being charged off. This pattern is consistent with the moderate path dependence detected by the Markov property test.

Lookback Window	Current	DPD_30	DPD_60	DPD_90+
3 months	0.144%	1.126%	5.883%	10.120%

Lookback Window	Current	DPD_30	DPD_60	DPD_90+
6 months	0.212%	1.120%	5.599%	9.163%
9 months	0.166%	0.972%	5.776%	9.475%
12 months (production)	0.198%	1.110%	6.282%	10.239%

Table 24: Lookback-Window Sensitivity of the 12-Month PD

The 12-month PD is robust to the choice of training window: across 3- to 12-month lookbacks, estimates range from 0.14% to 0.21% for Current and from 9.16% to 10.24% for DPD_90+, with no change in rank order.

5.6.2 Monte Carlo Validation

Check	Baseline	Adverse	Severely Adverse
Standard error / mean (2,000 paths)	0.43%	0.17%	0.09%
Running mean at 500 / 1,000 / 2,000 paths	$1.856M / $1.857M / $1.860M	$13.26M / $13.24M / $13.28M	$48.87M / $48.87M / $48.84M
Simulated vs. analytical expected defaults	225.9 vs. 226.0	—	—
Simulated vs. analytical expected loss	$1.860M vs. $1.848M (0.6%)	—	—

Table 25: Monte Carlo Convergence and Reconciliation

The simulation has converged in all scenarios (standard error below 0.5% of the mean), and the baseline reconciles to within 0.6% of the analytical expectation Σ EAD $\times$ P^{12} $\times$ LGD, confirming implementation accuracy.

5.6.3 Bayesian Validation

- **Cross-model consistency:** Bayesian expected loss ($2.11M) is within 13% of the Monte Carlo baseline ($1.86M); the difference reflects the behavioral uplift for accounts with weak or missing payments.
- **Monotonicity:** average Final PD increases strictly with delinquency severity (0.19%, 1.25%, 9.71%, 18.80%).
- **Flag separation:** High Score, Poor Behavior accounts are 3.8 times more likely to be past due than Normal accounts (22.3% versus 5.9%) and 6.1 times more likely to be 60+ days past due (4.1% versus 0.7%); Low Score, Excellent Behavior accounts have no past-due accounts.
- **Cap behavior:** the 85% posterior cap was not binding for any account, so results are not driven by the cap.

6. Discussion

6.1 Interpretation of Key Findings

The integrated framework produces four findings of practical significance. First, the Markov Chain, estimated from 926,155 transitions with narrow confidence intervals, projects gradual deterioration under stable conditions (Current share 62.52% → 59.43% over 12 months) and discriminates well out of sample (AUC 0.86 for default, 0.92 for serious delinquency).

Second, the Monte Carlo layer quantifies strong non-linear sensitivity to macroeconomic stress: expected loss rises from $1.86 million at baseline to $13.28 million under a mild recession and $48.84 million under a deep recession, with a 99th-percentile loss of $54.16 million. The baseline reconciles to the analytical expectation within 0.6%.

Third, the Bayesian component identifies 4,621 accounts ($91.1 million) whose credit scores understate their risk with this institution, with past-due rates 3.8 times those of normally behaving accounts, and a further 1,374 low-score members with excellent payment behavior. The priority tiers concentrate nearly two-thirds of flagged expected loss in 470 accounts.

Fourth, KL monitoring confirms that the calibrated transition matrix remains representative of the current portfolio; the only elevated signal is a February calendar effect, which the monitor correctly isolates.

6.2 Business Value and Practical Implications

The framework delivers five critical business outcomes for credit unions. First, real-time behavioral signals move beyond static credit scores to continuously updated risk signals derived from the credit union's own transaction data. Second, hidden risk detection identifies accounts whose risk is rising before it appears in standard score-based reports. Third, creditworthy member recovery surfaces members who carry low credit scores but demonstrate consistent on-time payments - candidates being missed by traditional underwriting. Fourth, precision-targeted interventions route the right member to the right program — collections outreach, hardship programs, or new lending opportunities. Fifth, account-level CECL provisioning replaces portfolio-wide averages with account-level expected loss estimates. Because every layer is persisted and time-stamped in Snowflake and governed by an explicit recalibration trigger, results are auditable and repeatable.

6.3 Directions for Future Research

Immediate priorities are to load transaction history for mortgage and line-of-credit products, to validate behavioral flags against subsequent delinquency on later snapshots, and to stress recovery rates alongside default frequency. Second-order or duration-dependent Markov models would address the detected path dependence, and time-varying transition matrices conditioned on unemployment and other macroeconomic covariates would address stationarity. Hidden Markov Models offer an extension accounting for unobserved latent borrower states. On the Bayesian side, a conjugate Beta-Binomial update would replace the piecewise multiplier structure with a formal posterior distribution. Replication across multiple credit unions remains the most critical step toward generalizability.

7. Conclusion

This research developed and validated an integrated credit risk framework combining Markov Chain migration modeling, Monte Carlo loss simulation, and Bayesian behavioral anomaly detection, governed

by KL divergence drift monitoring, for post-origination portfolio management in credit unions. Applied to 93,971 active loan accounts at Credit Union A observed from August 2025 to August 2026, the framework demonstrated both analytical rigor and practical utility.

The Markov Chain model provided precisely estimated transition probabilities and strong out-of-sample discrimination (AUC 0.86), projecting a gradual decline in Current accounts from 62.52% to 59.43% over 12 months. The Monte Carlo simulation quantified portfolio sensitivity to macroeconomic stress, with 12-month expected losses of $1.86 million at baseline, $13.28 million under a mild recession, and $48.84 million under a deep recession, reconciling to the analytical expectation within 0.6%. The Bayesian component identified 4,621 accounts ($91.1 million) whose credit scores understated their repayment risk - 3.8 times more likely to be past due than normally behaving accounts and 1,374 low-score members with excellent payment behavior. KL monitoring confirmed the model remains representative of the current portfolio.

For credit union practitioners, the framework provides a structured, interpretable, and CECL-compatible approach to four core post-origination questions: where the portfolio is migrating, how much it could lose under stress, which individual accounts require attention, and when the model itself needs to be recalibrated.

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